

Informe de experimento: results_validation_gpt-5.4_figure_multistep022

Generado el 2026-09-16 07:55 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio results_validation_gpt-5.4_figure_multistep022 (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_gpt-5.4_figure_multistep022
```

- Reports leídos: 1 (report.rescored.json usado en 0; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: react_nogate, pfagent
- Casos: case14
- Requests: archivo /private/tmp/claude-501/-Users-lzanda-Documents-REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMAR (1 report)
- Items por dificultad: multistep=2
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 07:55
- Costo total: 0.1861 USD (2 items; costo medio 0.09307 USD/item)
- Items totales: 2; ok=2; con error=0; tokens totales=66708
- Solved global: 0.0 % (0/2)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método react_nogate

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

User prompt (primer item) = request_text —242 caracteres; case=case14, seed=1083806780, k=1, request_id=case14-multistep-022-s0

Load case14, disconnect the line between bus 9 and bus 10, set the active load at bus 3 to 118.6 MW, set the active load at bus 4 to 53.6 MW, and finally run an N-1 contingency analysis ranked by max violations and report the 8 worst outages.

Método pfagent

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=False, memory=True

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3. Resultados agregados

Columnas del protocolo (medias por método $\times$ caso, del scoreboard_per_case), en los 4 grupos de metricas_pfagent_definicion: **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método $\times$ caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	\$
react_nogate	case14	1	0.0 (0/1)	3.02e-05	0.00	0.0 (0/1)	100.0 (1/1)	1.16e-04	100.0	–	5.0	28473	–	0.0 (0/1)	0.0807
pfagent	case14	1	0.0 (0/1)	3.90e-04	0.03	0.0 (0/1)	100.0 (1/1)	1.60e-03	100.0	–	5.0	38235	–	0.0 (0/1)	0.1054

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	multistep
react_nogate	0.0 (0/1)
pfagent	0.0 (0/1)

Solved por dificultad (%, items resueltos / items)

Method	multistep
react_nogate	0.0 (0/1)
pfagent	0.0 (0/1)

Verificación final (pfagent: V(x,c,z,y) aplicado a la respuesta final)

pass_first = pasó al primer intento; pass_retry = pasó tras el único reintento con el veredicto añadido; abstained = falló dos veces, respuesta forzada a fallo declarado sin números. N = items con final_gate activo (no todos los métodos lo tienen).

- Total losses: 17.4585 MW
- Voltage range: 1.0100 to 1.0900 pu

3. Violations, if any:

- Base case after your modifications has 7 voltage violations and 0 thermal violations.
- Voltage violations are overvoltage at buses: 1, 6, 7, 8, 9, 12, 13.

Top 8 N-1 contingencies ranked by max violations:

1. Outage 10-11 (line): converged, 8 voltage violations, 0 thermal violations, delta voltage violations = +1, worst Vm = 1.0100 pu, worst
 ↳ loading = 1.6473%

2. Outage 1-5 (line): converged, 7 voltage violations, 0 thermal violations, delta ...[1154 chars más]

--- Scoring ---

formulation=extra_arg_changes_result (exact=False); V_MAE=3.02e-05; F_MAE=0.00; faithful=100.0 %; solved=False (formulation); flags=-;

↳ error=-; tokens=28473; cost_usd=0.0807; wall=13.5 s

Método pfagent

Ejemplo: formulación fallida (formulation_exact=False)

request (case14-multistep-022-s0, case14, difficulty=multistep, seed=1083806780): Load case14, disconnect the line between bus 9 and bus
 ↳ 10, set the active load at bus 3 to 118.6 MW, set the active load at bus 4 to 53.6 MW, and finally run an N-1 contingency analysis

↳ ranked by max violations and report the 8 worst outages.

intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 9, "to_bus": 10}) -> modify_load({"bus_id": 3, "p_mw":
 ↳ 118.6}) -> modify_load({"bus_id": 4, "p_mw": 53.6}) -> run_n1_contingency({"top_k": 8, "criteria": "max_violations"})

executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 9, "to_bus": 10}) -> modify_load({"bus_id": 3, "p_mw":
 ↳ 118.6, "q_mvar": 0}) -> modify_load({"bus_id": 4, "p_mw": 53.6, "q_mvar": 0}) -> run_n1_contingency({"top_k": 8, "criteria":

↳ "max_violations", "max_candidates": 0})

fuentes de la traza: archivo de traza completo

--- Ronda 1 ---

tool: load_case({"case_name": "case14"})

output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 264.0857422973543,

↳ "total_gen_capacity_mw": 772.4000000000001}

gate: n/a (no es un resultado de flujo)

--- Ronda 2 ---

tool: disconnect_line({"from_bus": 9, "to_bus": 10})

output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,

↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -5.032452347367612, "is_violation": false,

↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0449999999999997, "va_deg": -5.627497640283712, "is_violation": false,

gate: PASS (reasons=-, mismatch=6.74e-09 MW)

--- Ronda 3 ---

tool: modify_load({"bus_id": 3, "p_mw": 118.6, "q_mvar": 0})

output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,

↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.627497640283712, "is_violation": false,

↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0449999999999997, "va_deg": -5.627497640283712, "is_violation": false,

gate: PASS (reasons=-, mismatch=7.38e-09 MW)

--- Ronda 4 ---

tool: modify_load({"bus_id": 4, "p_mw": 53.6, "q_mvar": 0})

output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,

↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000004, "va_deg": -5.67478369579074, "is_violation": false,

↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0449999999999997, "va_deg": -5.627497640283712, "is_violation": false,

gate: PASS (reasons=-, mismatch=9.86e-09 MW)

--- Ronda 5 ---

tool: run_n1_contingency({"top_k": 8, "criteria": "max_violations", "max_candidates": 0})

output: {"plot_type": "n1_ranking", "figure_json":

↳ {"data": [{"cliponaxis": false, "hoverinfo": "text", "hovertext": [{"line": "10-11", "vm_pu": 1.01, "va_deg": 0.0, "is_violation": true, "violation_type": "overvoltage"}], "text": "10-11", "vm_pu": 1.01, "va_deg": 0.0, "is_violation": true, "violation_type": "overvoltage"}], "xaxis": "line", "yaxis": "vm_pu"}]

↳ [{"line": "10-11", "vm_pu": 1.01, "va_deg": 0.0, "is_violation": true, "violation_type": "overvoltage"}], "xaxis": "line", "yaxis": "vm_pu"}]

↳ [19572 chars más]

gate: n/a (no es un resultado de flujo)

--- Ronda 6 ---

LLM: 1. **Summary:** I loaded IEEE 14-bus, disconnected line **9-10**, set the active loads at **bus 3 = 118.6 MW** and **bus 4 = 53.6 MW**

↳ (interpreted as setting **Q = 0 MVar** since only active load was specified), and then ran an **N-1 contingency analysis** ranked by
 ↳ **maximum violations**, reporting ...[1880 chars más]

--- Respuesta final ---

1. **Summary:** I loaded IEEE 14-bus, disconnected line **9-10**, set the active loads at **bus 3 = 118.6 MW** and **bus 4 = 53.6 MW**
 ↳ (interpreted as setting **Q = 0 MVar** since only active load was specified), and then ran an **N-1 contingency analysis** ranked by
 ↳ **maximum violations**, reporting the **8 worst outages**.

↳ **maximum violations**, reporting the **8 worst outages**.

2. **Key numbers:**

- **Total load:** **286.7497 MW**

- **Total generation:** **304.2194 MW**

- **Total losses:** **17.4697 MW**

- **Voltage range:** **1.0100 pu** to **1.0900 pu**

3. **Violations:**

- **Base case after your modifications converged.**

- **Base-case voltage violations:** **7 overvoltage violations** at buses **1, 6, 7, 8, 9, 12, 13**

- **Base-case thermal violations:** **none**

Top 8 N-1 outages ranked by max violations:

1. **Outage** ...[1380 chars más]

--- Scoring ---

formulation=extra_arg_changes_result (exact=False); V_MAE=3.90e-04; F_MAE=0.03; faithful=100.0 %; solved=False (formulation); flags=-;

↳ error=-; tokens=38235; cost_usd=0.1054; wall=29.9 s

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- $V_MAE \geq 1e-4$ p.u. en 1 filas método×caso con solver: `pfagent/case14` (3.90e-04).
- El gate verificó 6 resultados de flujo, 0 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved más alto: `pfagent` 0.0 % (0/1); más bajo: `react_nogate` 0.0 % (0/1).
- Formulación exacta agregada sobre métodos con tools: 0.0 % (0/2).
- Errores de formulación más frecuentes: `extra_arg_changes_result=2`.
- Ningún item reclamó éxito sobre un fallo (`claimed_success_on_failure=0`).
- 0 de 2 items con mutación de red reportaron estado obsoleto (`stale_state`).
- Trazabilidad de números (Faith.) media con tools: 100.0 %.
- Costo total 0.1861 USD; el método más caro fue `pfagent` con 0.1054 USD (57 %).
- Métodos con tools: 5.00 tool calls medias por item (máximo 5).

6. Archivos

Rutas absolutas bajo `results_validation_gpt-5.4_figure_multistep022.report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_validation_gpt-5.4_figure_multistep022/report.json
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_figure_multistep022/scoreboard.json
↳ /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_figure_multistep022/scoreboard.csv
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_figure_multistep022/scoreboard.md
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_figure_multistep022/scoreboard_per_case.json
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_figure_multistep022/scoreboard_per_case.json
traces (2 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_figure_multistep022/traces
```